ETH zürich

KOF

Principles of Macroeconomics

Lecture 14: Exam Preparation

Prof. Dr. Jan-Egbert Sturm

Schedule

Date	Lecture	Topic	Chapters
16/09/2025	1	Introduction	1, 2
23/09/2025	2	National Accounting	20
30/09/2025	3	Production and Growth	21
07/10/2025	4	Unemployment and the labour market	22
14/10/2025	5	Saving, investment and the financial system	23
21/10/2025	6	The Monetary System	24
28/10/2025	7	International Trade	3,15,17
04/11/2025	8	Open-Economy Macroeconomics	25
11/11/2025	9	Business Cycles	26
18/11/2025	10	Keynesian Economics and IS-LM Analysis	27
25/11/2025	11	Aggregate Demand and Aggregate Supply	28
02/12/2025	12	Monetary and Fiscal Policy	29
09/12/2025	13	Phillips Curve / Supply-side policies	30,31
16/12/2025	14	Exam preparation	

Final Exam, End-of-semester examination: Tuesday, **13.01.2026** at **14:00** (CET)

- Duration, content and structure
  - Duration of the exam is ***90 minutes***
    - Exam begins at 14:00
      (please be there at **13:30**)
  - Content covers all the material discussed in
    - Class,
    - Problem sets,
    - Textbook (corresponding chapters).
  - Structure of the exam:
    - Single-Choice questions
    - Multiple-Choice questions
    - Open questions
      (see Mock Exam on Moodle)

- Allowed material:
  - Non-programmable calculator without graphics function
  - Non-electronic dictionary
- Required material:
  - **Student identity card**
  - Dark pen or ballpen (but no pencil)
  - Ideally, one or two additional colours for graphs
- Room:
  - check on Moodle and your ETH e-mail
- If needed, we will publish additional important exam information on Moodle
  (please check the day before the exam)

End-of-Semester **Repetition examination**: Monday, **23.02.2026** at **14:00** (CET)

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Only students who did **not** pass the Final Exam are eligible to take the Repetition Exam.
Students are **required to register** with the Examination Office (observe the deadline)

- Duration, content and structure
  - Duration of the exam is ***90 minutes***
    - Exam begins at 14:00 (please be there at **13:40**)
  - Content covers all the material discussed in
    - Class,
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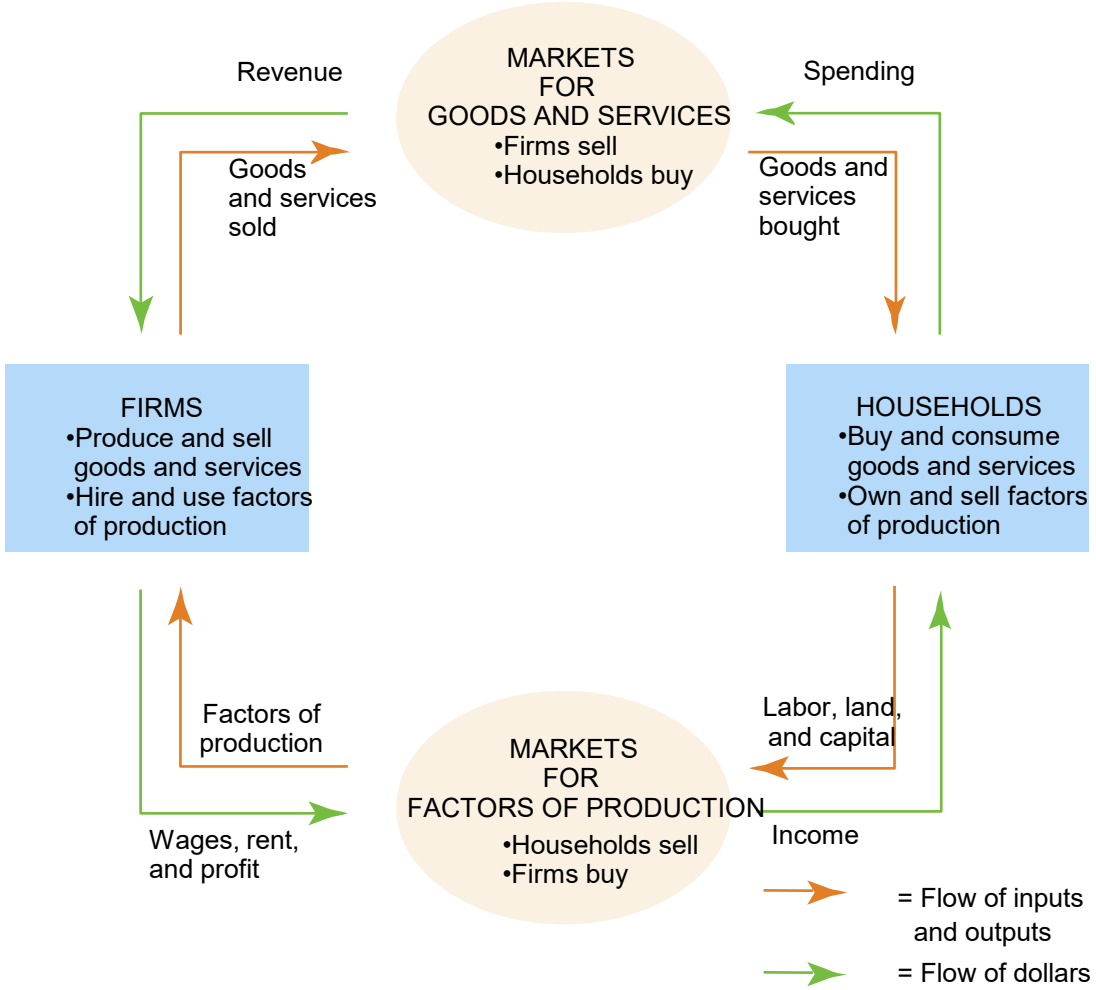
Lecture 14: Exam preparation

Prof. Dr. Jan-Egbert Sturm

The Circular Flow



National accounting

- ***GDP*** is the total market value of all final goods and services produced within a country in a given period of time. GDP (Y) is the sum of the following: $Y = C + I + G + NX$
  - Consumption (C)
  - Investment (I)
  - Government Purchases (G)
  - Net Exports ($NX = X - M$)
- ***Real GDP*** values the production of goods and services at constant prices.
- ***Nominal GDP*** values the production of goods and services at current prices.
- The ***GDP deflator*** is the ratio of the two: $GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$
- The ***consumer price index*** (CPI) is a measure of the overall cost of the goods and services bought by a typical consumer: $CPI_t = \frac{Price\ of\ basket\ in\ period\ t}{Price\ of\ basket\ in\ base\ period} \times 100$
- (Consumer) ***inflation*** is the growth in CPI: $\pi_t = \frac{CPI_t}{CPI_{t-1}} \times 100$

Productivity and Growth

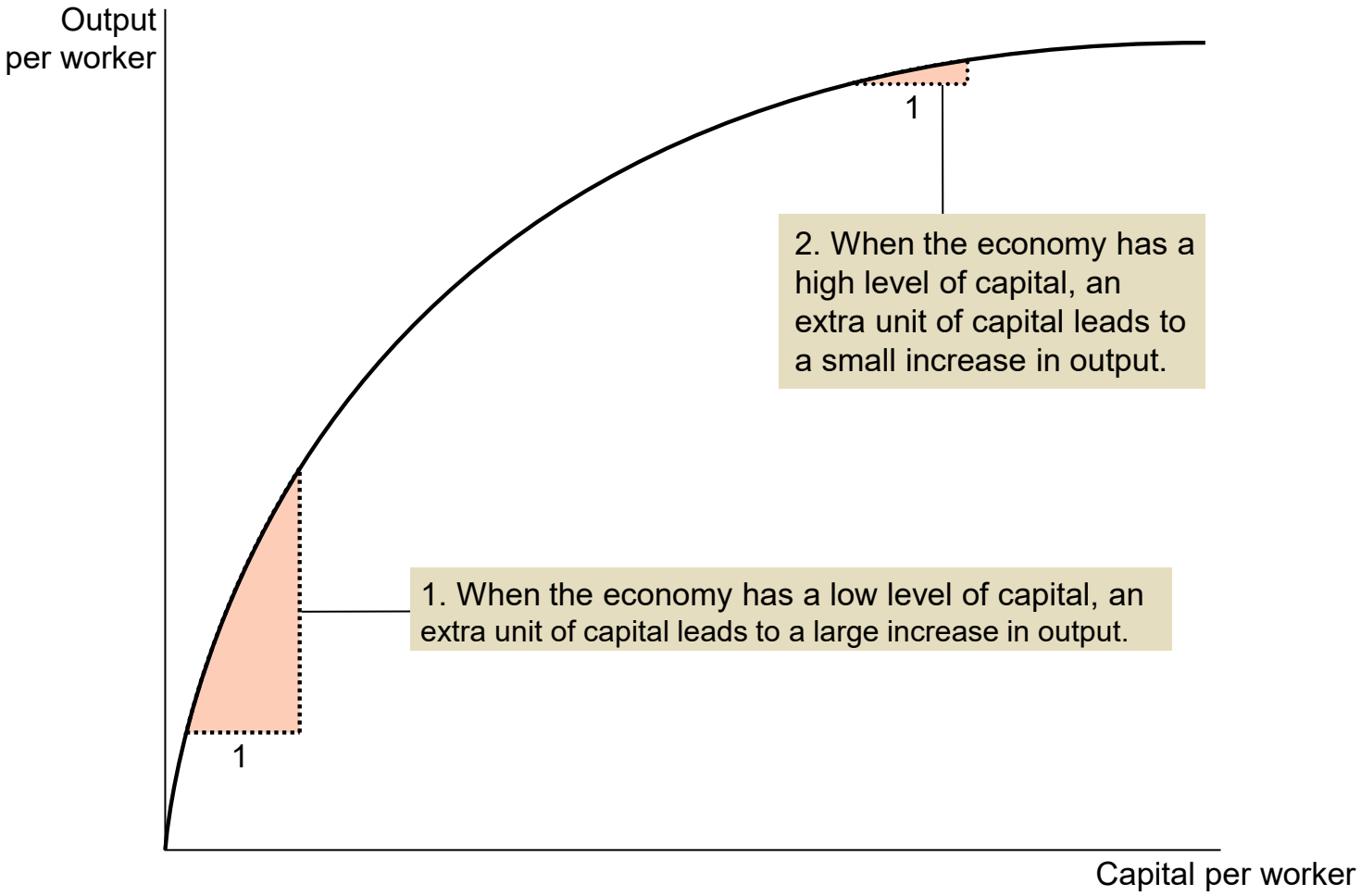
- Labour productivity (Y/L) refers to the amount of goods and services produced from each unit of labour input (L)
- Factors of production determine productivity:
  - Physical capital (K)
  - Human capital (H)
  - Natural resources (N)
  - Technological knowledge (A)

Production Function

- $Y = A\,F(L, K, H, N)$
  - $F(\,)$ is a function that shows how the inputs are combined.
- Factors that affect growth:
  - investment (also from abroad),
  - education,
  - health and nutrition,
  - property rights and political stability,
  - free trade,
  - research and development,
  - population growth

Illustrating the Production Function

- As the stock of capital rises, the extra output produced from an additional unit of capital falls; this property is called ***diminishing returns***.
- Because of diminishing returns, an increase in the saving rate leads to higher growth only for a while.



Saving and Investment

- The financial system consists of the group of institutions in the economy that help to match one person's saving with another person's investment.
- Financial Markets
  - Stock Market / Bond Market
- Financial Intermediaries
  - Banks / Mutual Funds
- $S = I$
  - We summarize all these assets into "loanable funds"
    - Loanable funds refers to all income that people have chosen to save and lend out, rather than use for their own consumption
  - Supply of loanable funds (therefore) comes from (private & public) savings
  - Demand for loanable funds comes from (domestic and foreign) investment
  - The equilibrium of the supply and demand for loanable funds determines the real interest rate

Government Saving

- Surplus and Deficit
  - If $T > G$, the government runs a budget surplus
    - The surplus of $T - G$ represents public saving
  - If $G > T$, the government runs a budget deficit

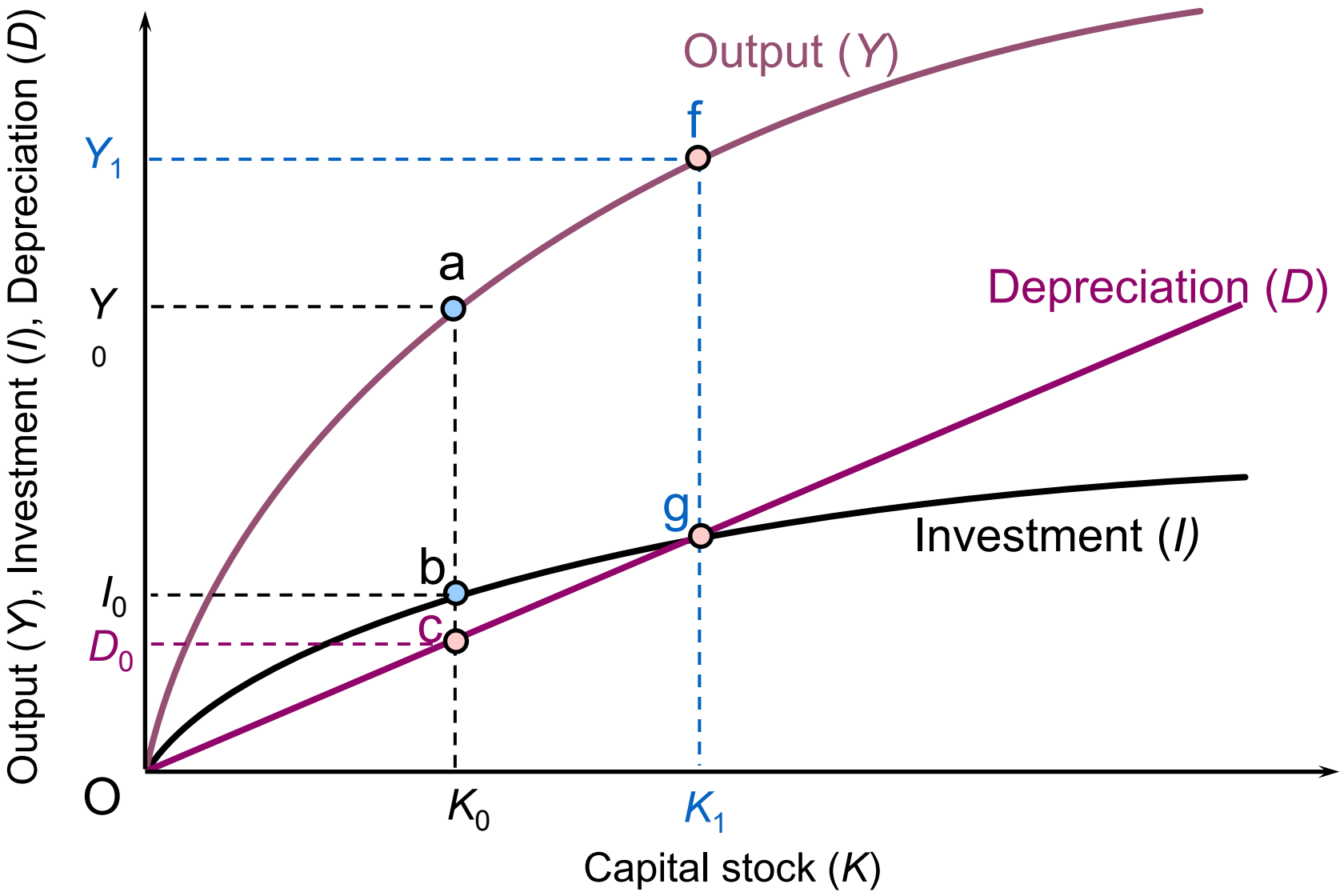
- Question: Assume a closed economy. Consumption spending= 350 bn, net taxes=270 bn, household saving=250 bn, investment spending= 220 bn. What is the government budget deficit or surplus?
  - Recall: closed economy implies $S = I$
  - $S =$ household saving (250) + government saving (x), $I =$ investment spending (220)
  - $250 + x = 220 \rightarrow x = -30$
  - Government deficit is 30 bn

Solow Model

- Long-run growth is exogenously determined

- economy will always converge towards a steady state rate of growth, which depends only on the rate of technological progress and the rate of labour force growth.
- In steady state, output growth per capita is nil
- Only continuing technological progress can explain continuing growth in Y.

Investment, Growth, and Output

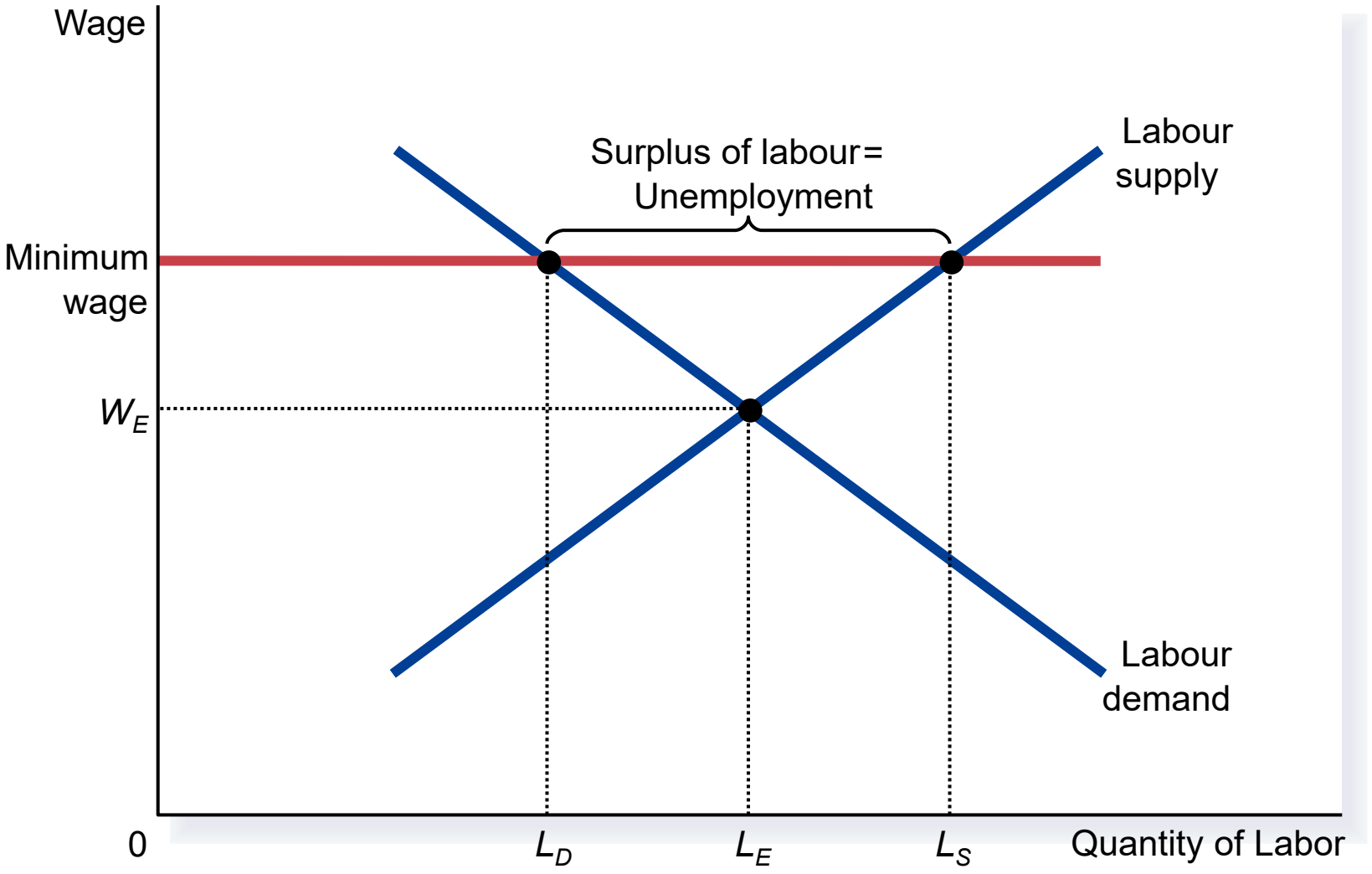


Unemployment

- The ***natural rate of unemployment*** is unemployment that does not go away on its own even in the long run
- ***Cyclical unemployment*** refers to the year-to-year fluctuations in unemployment around its natural rate
- The ***labour force*** is the total number of workers, including both the employed and the unemployed
- ***Frictional unemployment*** refers to the unemployment that results from the time that it takes to match workers with jobs
- ***Structural unemployment*** is the unemployment that results because the number of jobs available in some labour markets is insufficient to provide a job for everyone who wants one

- $Unemployment\ rate = \frac{Number\ unemployed}{Labour\ force} \times 100$
- $Labour\ force\ participation\ rate = \frac{Labour\ force}{Adult\ population} \times 100$

Unemployment from a Wage Above the Equilibrium Level

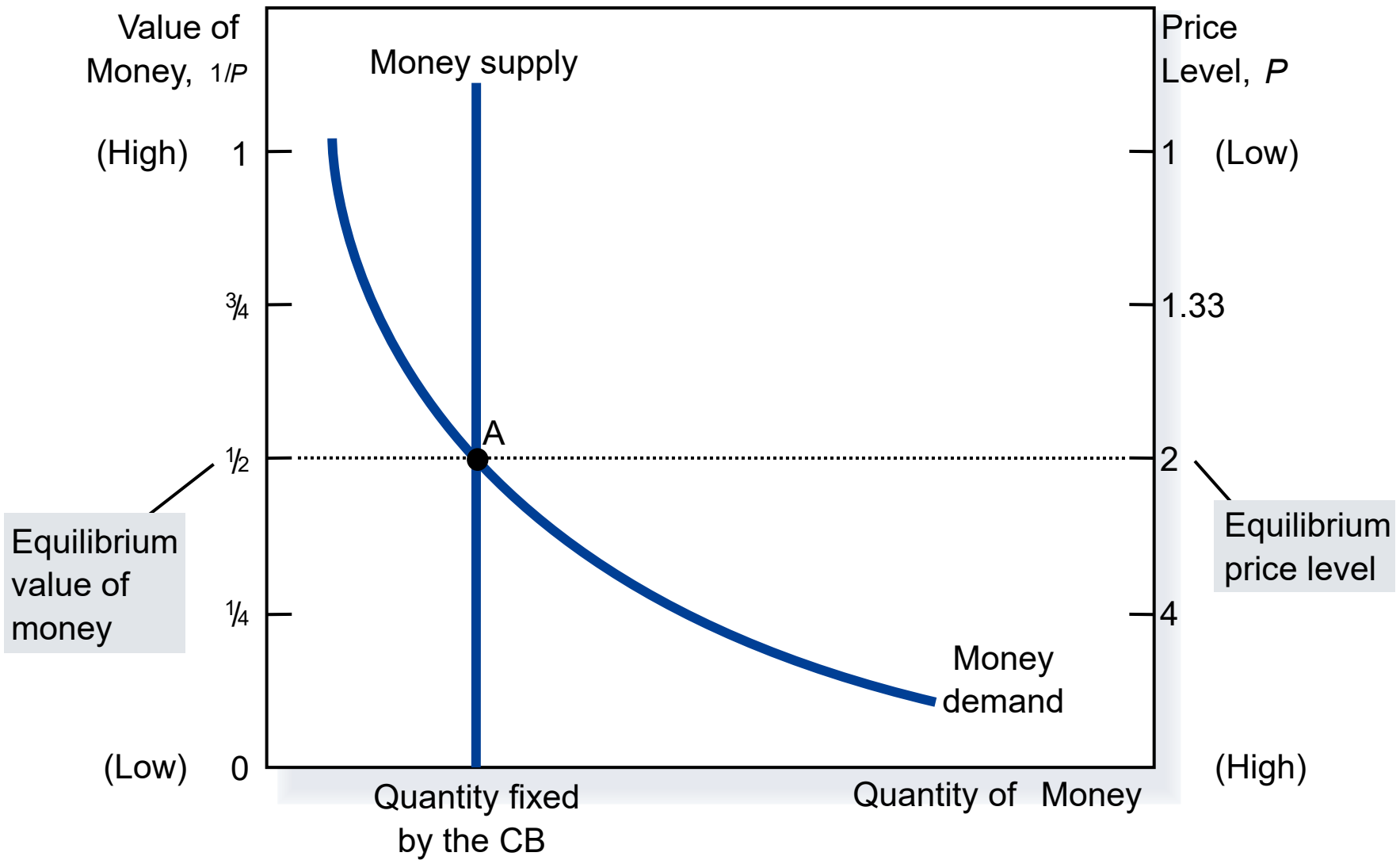


The Monetary System

- ***Money*** is the set of assets in an economy that people regularly use to buy goods and services from other people
- ***Liquidity*** is the ease with which an asset can be converted into the economy's medium of exchange
- Two Primary Functions of Central Banks
  - Act as a banker's bank, making loans to banks and as a lender of last resort
  - Conducts monetary policy (by controlling the money supply)
    - Influencing (controlling) the internal value of the currency (price stability)
    - Influencing (controlling) the external value of the currency (exchange rate)

- The ***reserve ratio*** is the fraction of deposits that banks hold as reserves
- The ***money multiplier*** is the amount of money the banking system generates with each dollar of reserves. $Money\ Multiplier\ = 1/R$ (where R is the reserve ratio)
- ***Inflation*** is an increase in the overall level of prices

How the Supply and Demand for Money Determine the Equilibrium Price Level



Inflation

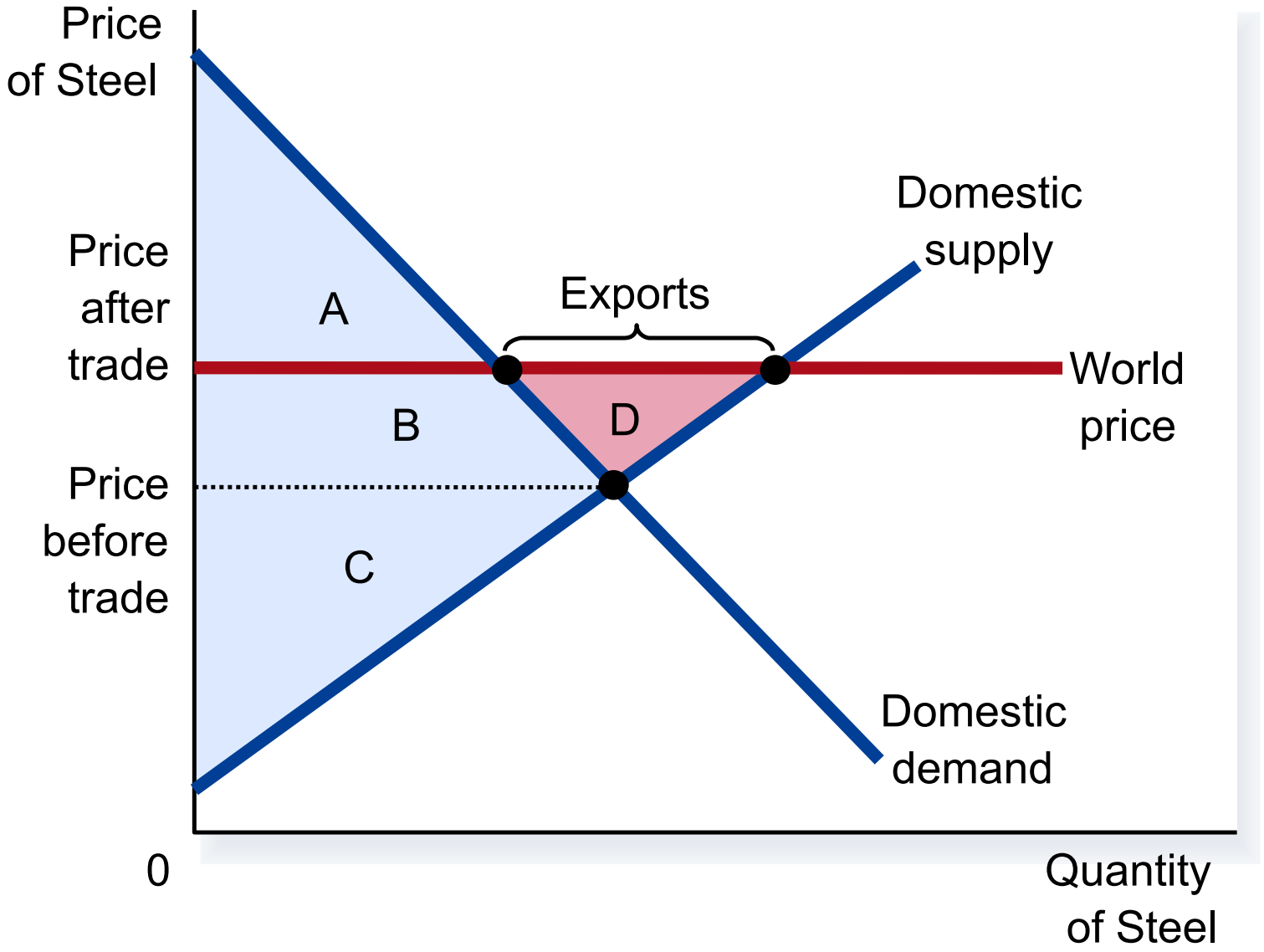
- The ***Fisher effect*** refers to a one-to-one adjustment of the nominal interest rate to the inflation rate
- According to the Fisher effect, when the rate of inflation rises, the nominal interest rate rises by the same amount
- The real interest rate stays the same
  - It is determined on the loanable funds market

- Costs of inflation:
  1. Shoe leather costs
  2. Menu costs
  3. Relative price variability
  4. Tax distortions
  5. Confusion and inconvenience
  6. Arbitrary redistribution of wealth

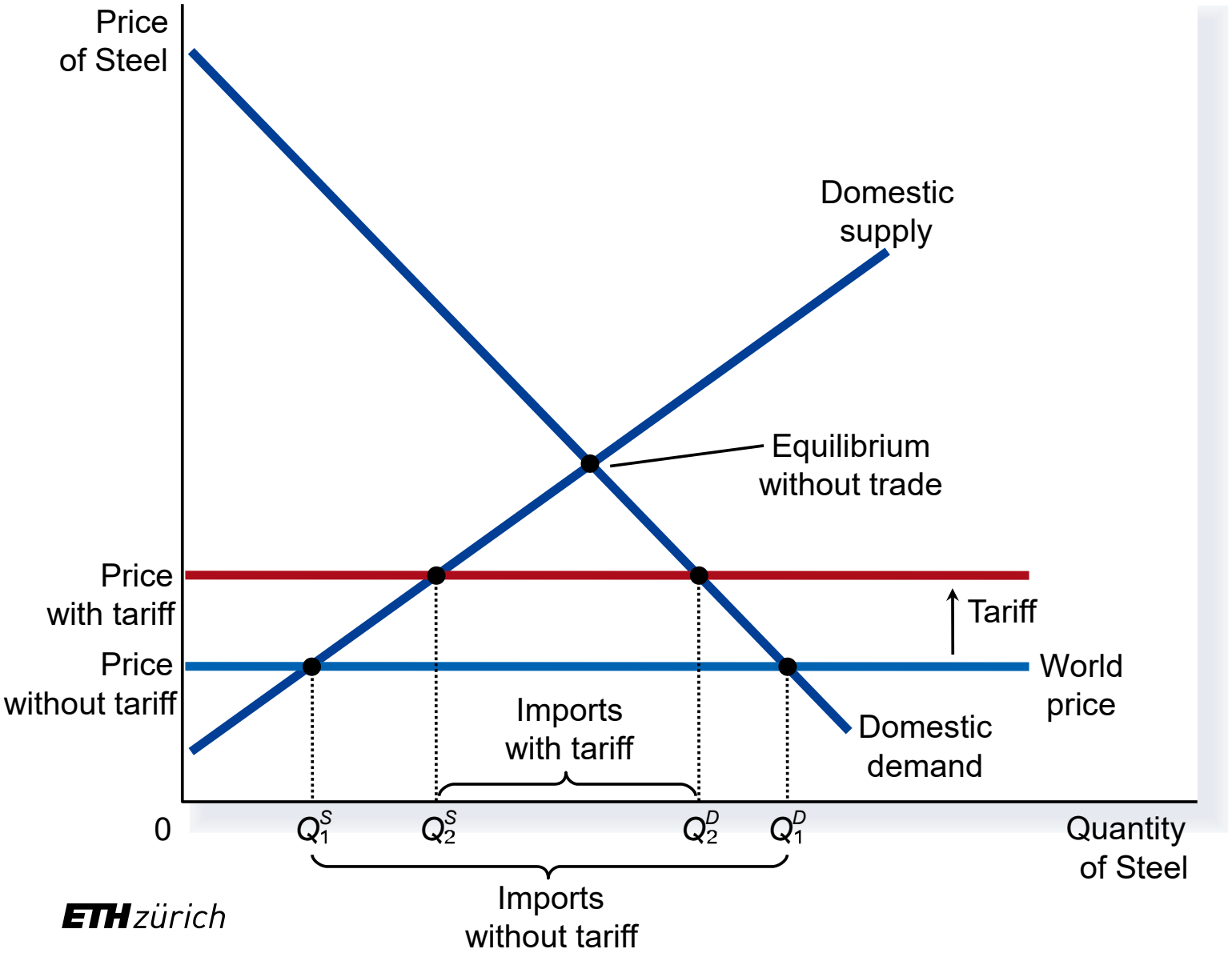
International Trade

- ***Absolute advantage***: The comparison among producers of a good according to their productivity
- ***Comparative advantage***: Compares producers of a good according to their ***opportunity cost***
  - The producer who has the smaller opportunity cost of producing a good is said to have a ***comparative advantage*** in producing that good
- Benefits of Trade
  - Trade can (ultimately) benefit everyone in a society because it allows people to specialize in activities in which they have a comparative advantage

How Free Trade Affects Welfare in an Exporting Country



The Effects of a Tariff

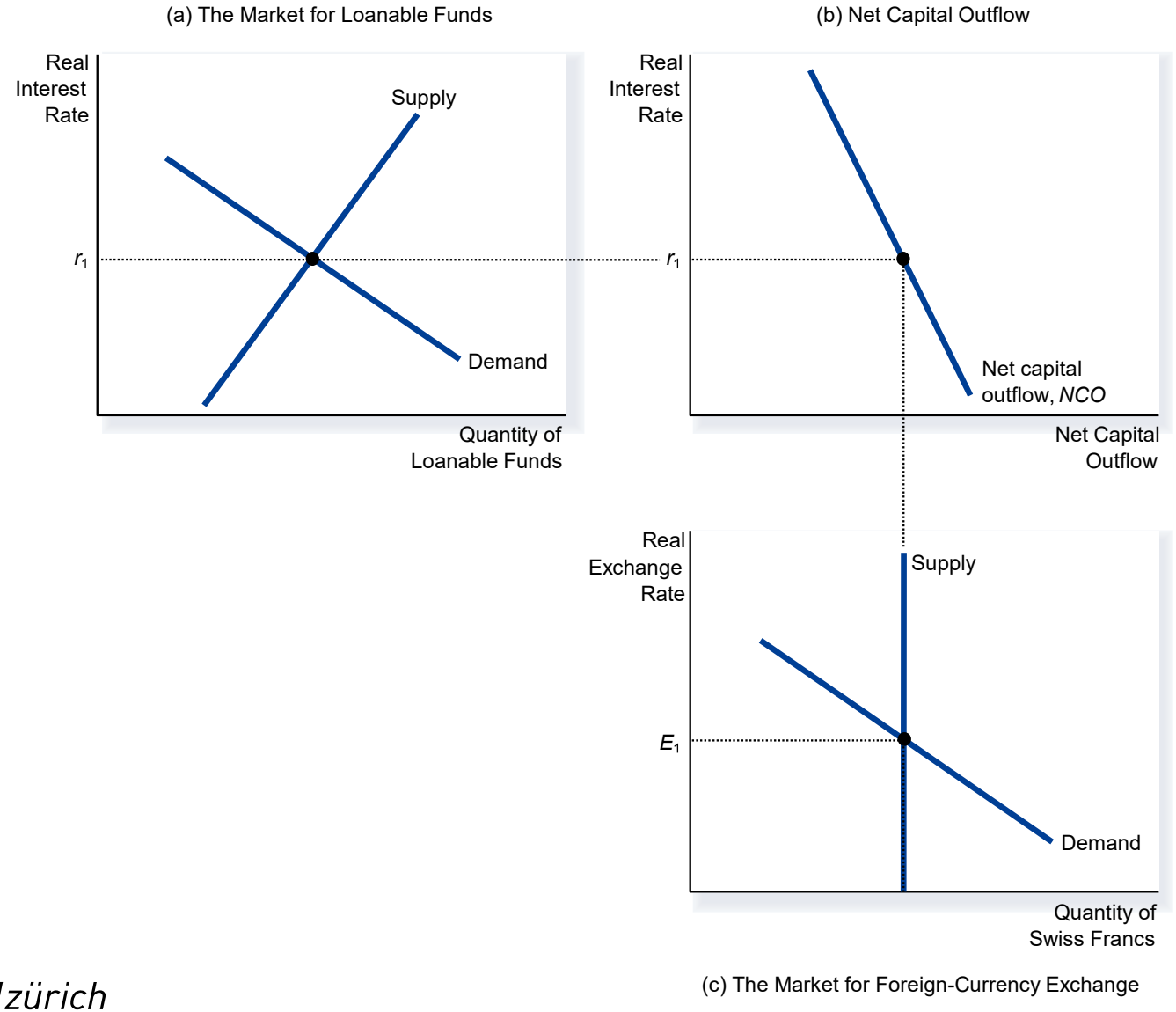


- A ***tariff*** is a tax on goods produced abroad and sold domestically.
- Tariffs raise the price of imported goods above the world price by the amount of the tariff
- An ***import quota*** is a limit on the quantity of a good that can be produced abroad and sold domestically

Open Economy Macro

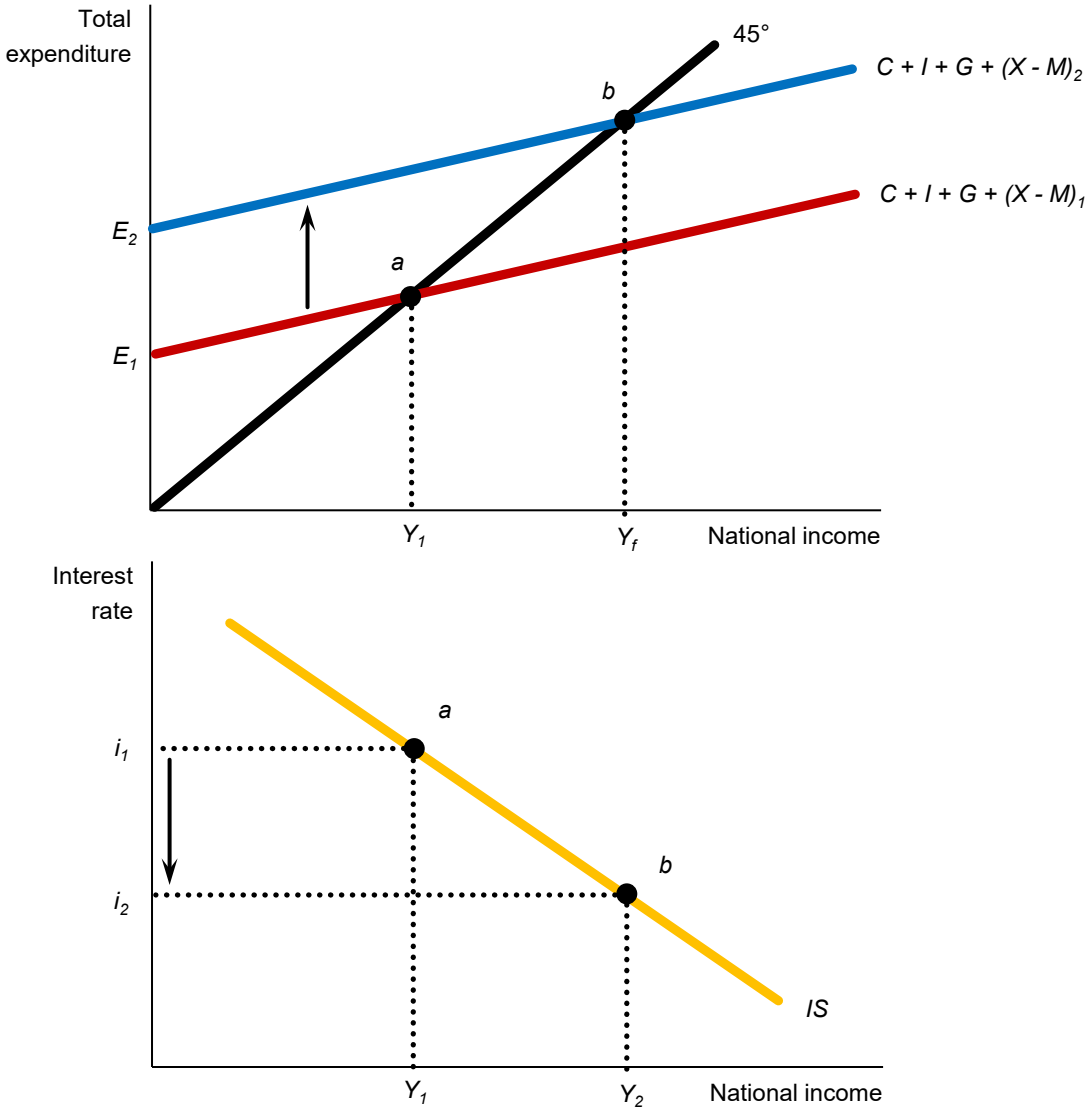
- ***Net exports*** (NX) are the value of a nation's exports minus the value of its imports
  - trade deficit: Imports > Exports
  - trade surplus: Exports > Imports
- ***Net capital outflow*** (NCO) refers to the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreigners
- $NCO = NX$
- The ***nominal exchange rate*** is the rate at which a person can trade the currency of one country for the currency of another.
  - Appreciation refers to an increase in the value of a currency as measured by the amount of foreign currency it can buy.
  - Depreciation refers to a decrease in the value of a currency as measured by the amount of foreign currency it can buy.
- The ***real exchange rate*** is the rate at which a person can trade the goods and services of one country for the goods and services of another: $Real\ exchange\ rate = \frac{Nominal\ exchange\ rate \times Domestic\ price\ level}{Foreign\ price\ level}$

The Real Equilibrium in an Open Economy



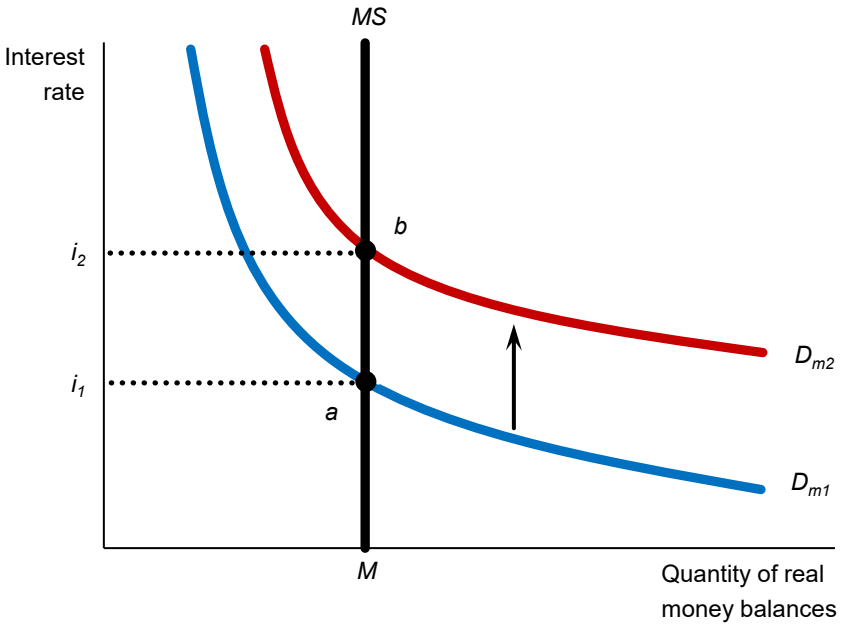
The IS Curve

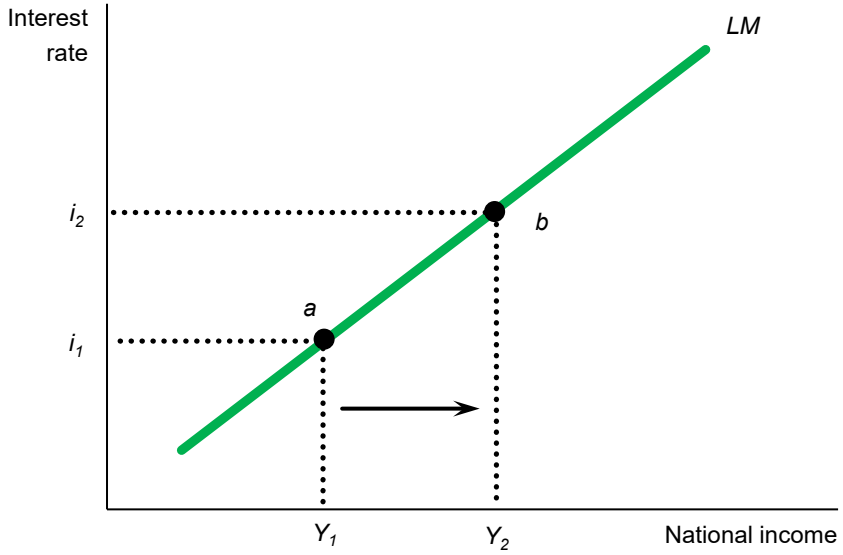
- The IS curve is derived from the Keynesian cross diagram
- In the upper panel, the initial equilibrium is point a
- A fall in the interest rate raises the expenditure line and a new equilibrium occurs at point b



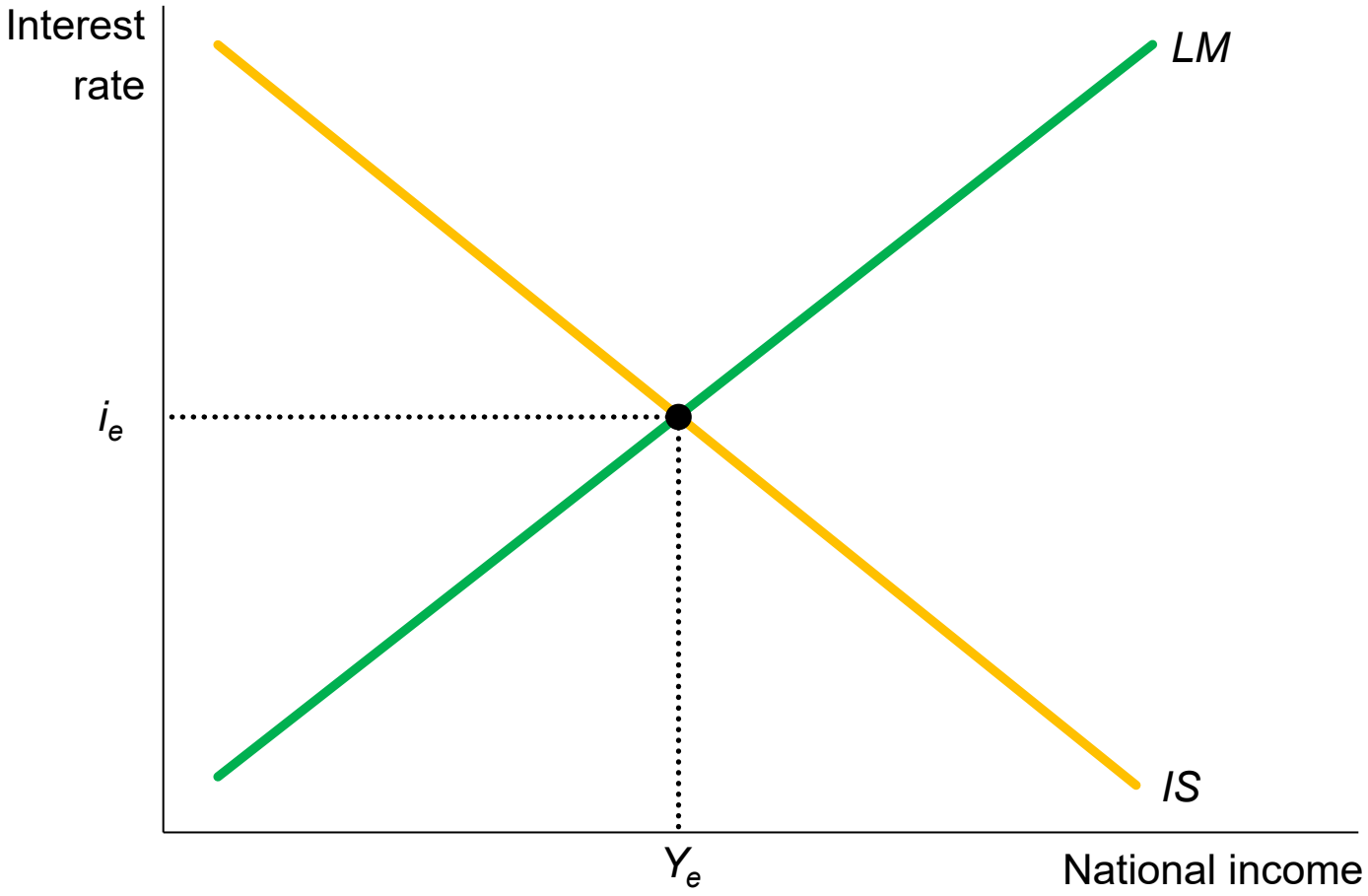
The LM Curve

- The LM curve is derived from our money market diagram
- In panel (a) initial equilibrium is point a
- A rise in national income increases the demand for money and a new equilibrium occurs at point b





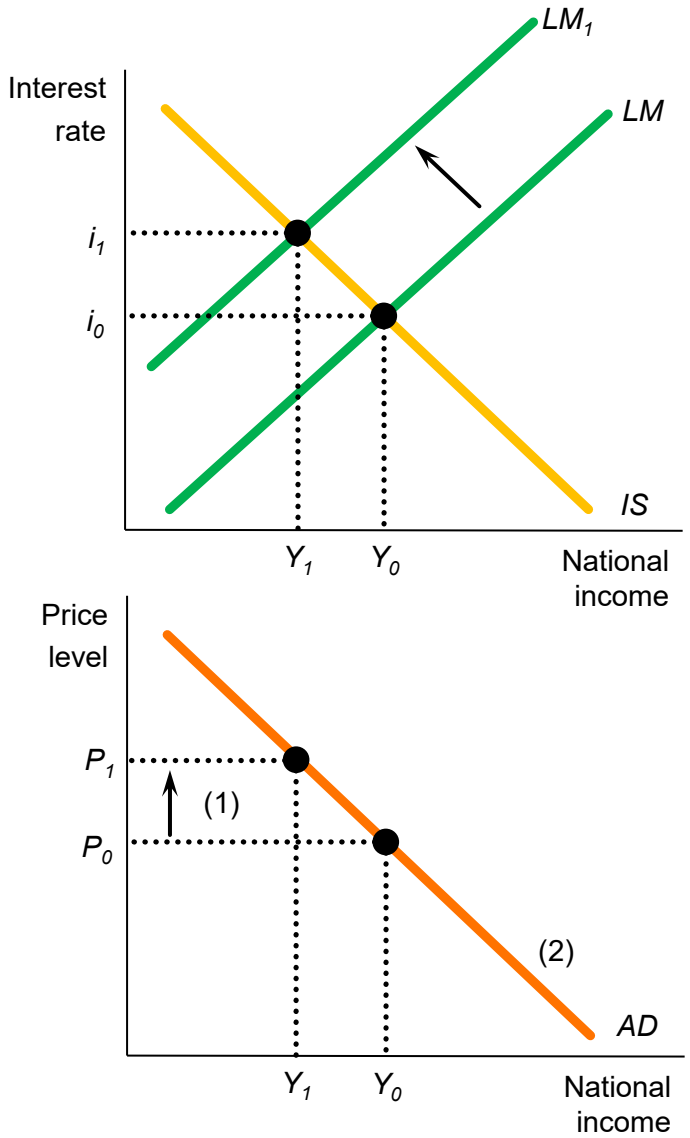
General equilibrium in the IS/LM Model



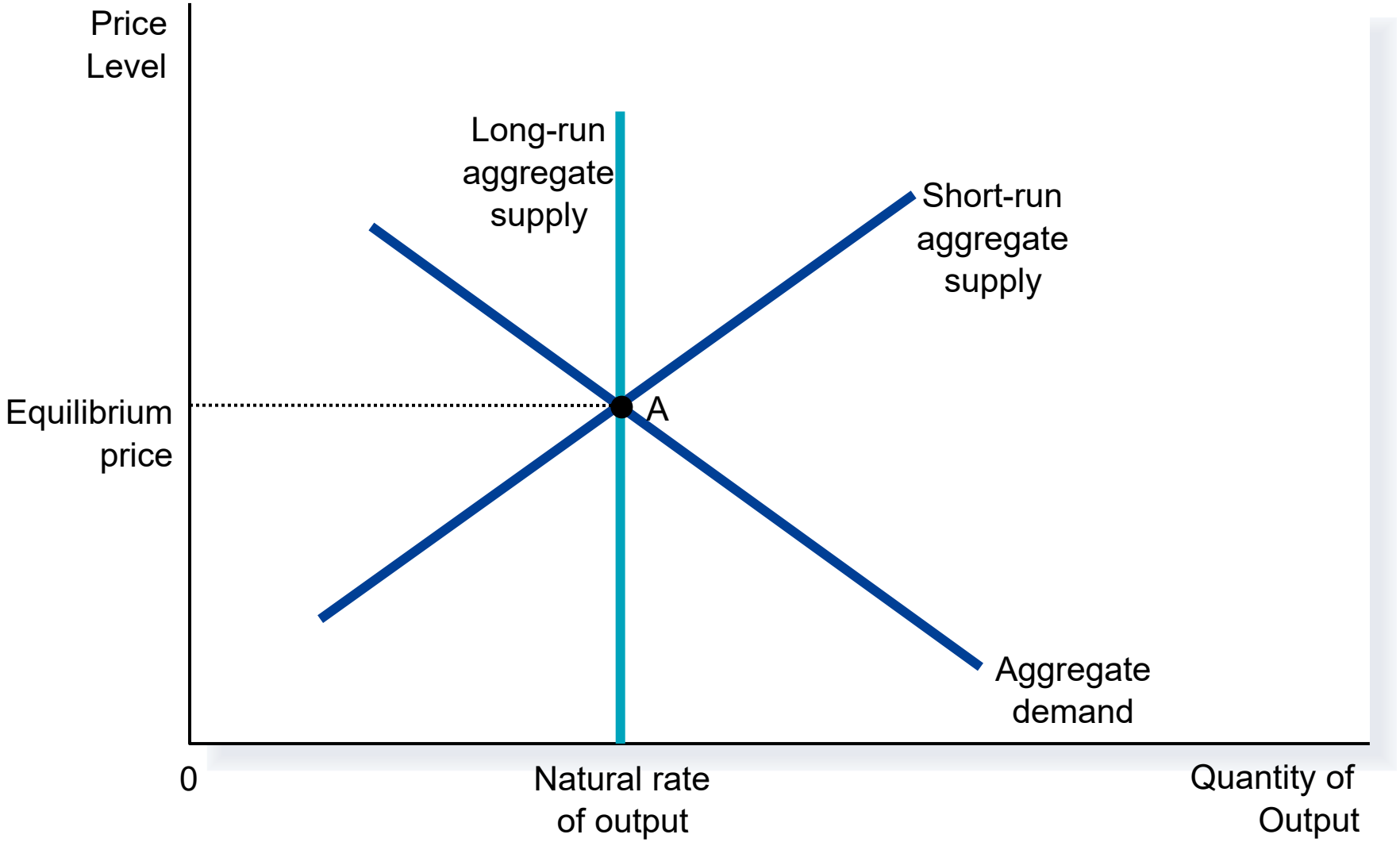
From the IS/LM Model to the AD curve

- A rise in the price level reduces real money balances and shifts the LM curve to the left
- This raises the equilibrium interest rate and lowers national income

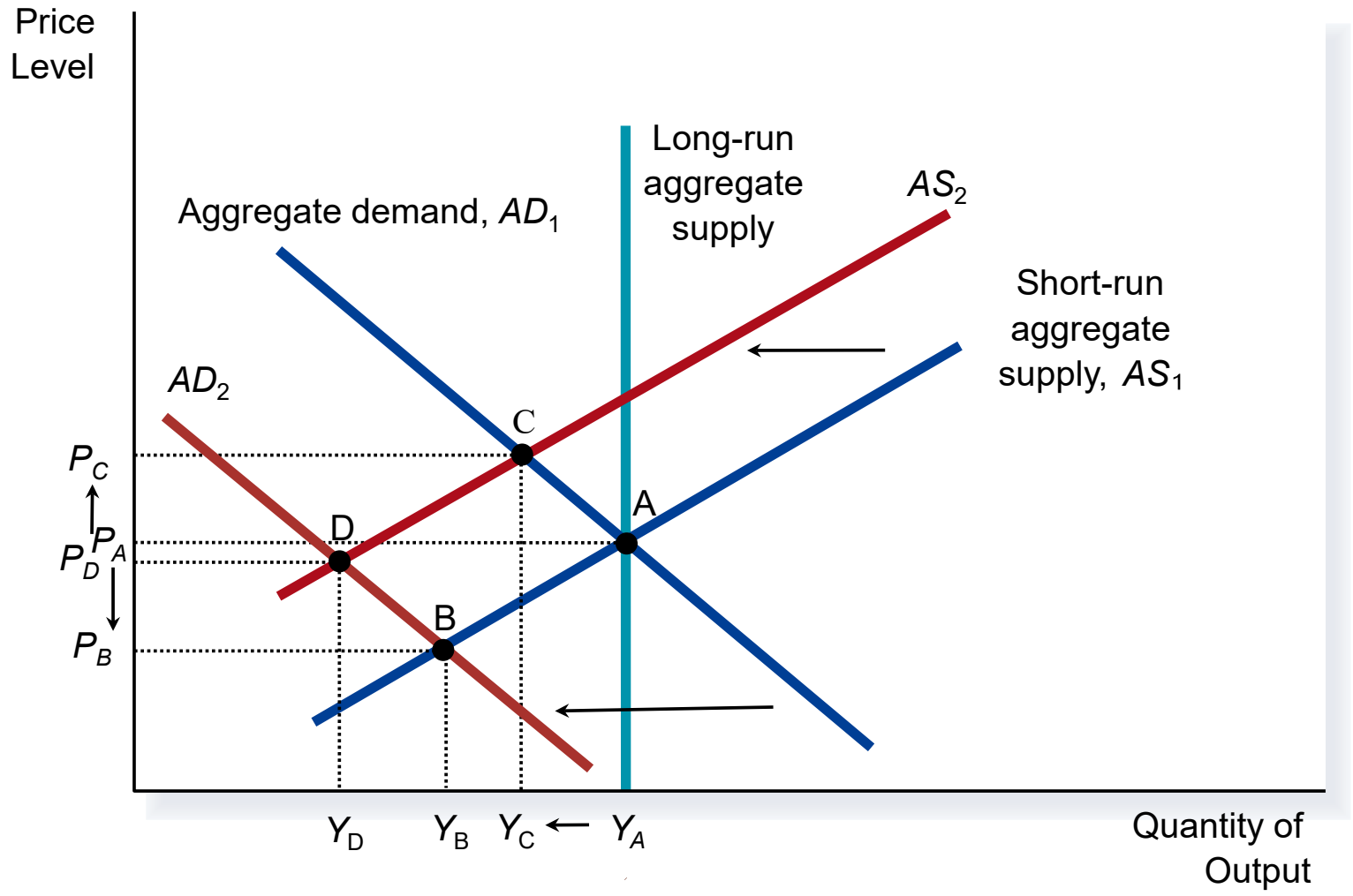
- The aggregate demand curve AD shows the inverse relationship between the price level and national income



Aggregate Demand and Aggregate Supply



The Economic Side of the Corona Crisis in the AD/AS Framework



Aggregate Demand and Aggregate Supply

- Aggregate demand (stemming from IS & LM)
  - $Y = C + I + G + NX$ & $M/P = L(i, Y)$

- The ***multiplier effect*** refers to the additional shifts in aggregate demand that result when expansionary fiscal policy increases income and thereby increases consumer spending
- The reduction in demand that results when a fiscal expansion raises the interest rate is called the ***crowding-out effect***

- Aggregate supply
  - $Y = \hat{Y} + a\,(P - Pe)$

Phillips Curve

- The ***Phillips curve*** shows the short-run trade-off between inflation and unemployment
  - The long-run Phillips curve is vertical at the natural rate of unemployment
  - Monetary policy could be effective in the short run, but not in the long run

- ***Okun's Law*** states that a one-percent decrease in unemployment is associated with roughly two percentage points of additional growth in real GDP
- The ***sacrifice ratio*** is the number of percentage points of annual output that is lost in the process of reducing inflation by one percentage point